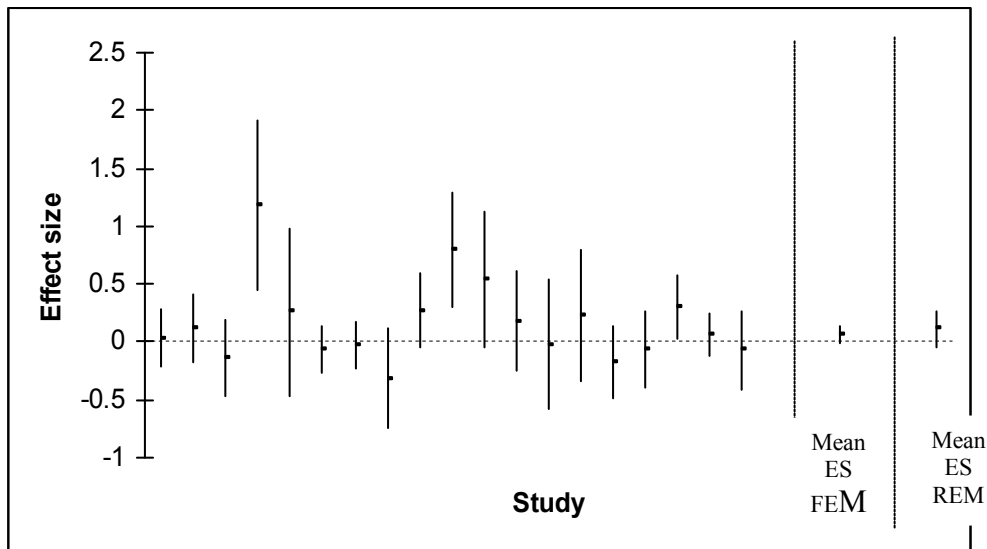


Information about the oral exam

For the exam, the paper is the most important. 10 of the 20 points are assigned to the paper, and additionally you will also get during the oral exam one or more questions regarding the paper. Besides, you probably will get one or more questions regarding the remaining of the course. Keep in mind that you do not know the textbook or slides by heart. On the contrary, the exam is an open book exam, but you will be tested if you understand what is discussed in the textbook and/or in the classes.

Example exam questions

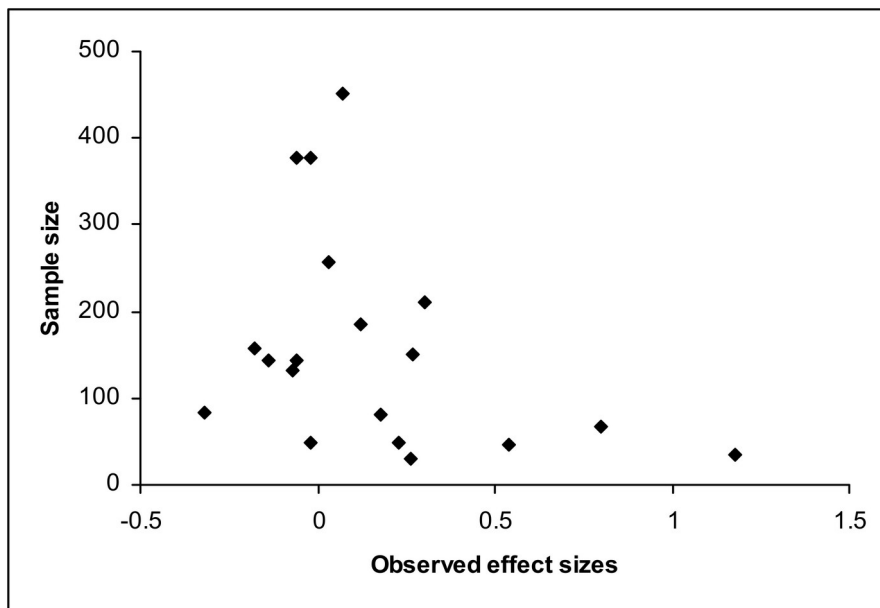
1. Below, you can see a graph with the 95 % confidence intervals for the observed and averaged effect sizes from a meta-analysis from Raudenbush (1984) regarding the teacher expectancy effect on the scores from pupils. What can you learn from it?



A possible answer:

- *there are 19 studies that are combined*
- *most of the observed effect sizes are positive, but are statistically not significant at the .05 significance level*
- *the combined effect is about .1 or .2. When using the fixed effects model, the combined effect size estimate is possibly statistically significant at the .05 level. When using the random effects model, the combined effect size estimate is somewhat larger, but the estimated accuracy is somewhat smaller (the confidence interval is larger) and the average effect turned out to be statistically not significant.*

2. Below, you can see a graph from effect sizes that were collected in a meta-analysis from Raudenbush (1984) regarding the teacher expectancy effect on the scores from pupils. What can you learn from it?



A possible answer:

- *there are 19 effect sizes that are combined*
- *as you would expect, the variation in observed effect sizes is smaller for larger studies*
- *there is a relation between the observed effect sizes and the sample size, in that on average the observed effect sizes are on average smaller for larger studies as compared to smaller studies.*
- *a possible explanation for this deviation from a symmetrical 'funnel' is that there is some bias, for instance publication or reporting bias: negative study outcomes, that are more likely to be observed in smaller studies, may have remained unpublished.*

3. Explain: "In a mixed effects meta-analysis, we model three sources of variation: sampling variation, moderating effects and residual study heterogeneity."

A possible answer:

- *in a FEM, we assume that all studies are estimating the same effect, such that there is only one source of variation: sampling variation since in primary studies typically only a sampling instead of the whole population is investigated*
- *the model can be extended by allowing that the population effects vary randomly (a REM)*
- *another possible extension is the inclusion of moderator variables, that describe how population effects vary according to study characteristics*
- *a combination of both extensions is the MEM: it assumes that the population effects may vary partly randomly, partly according to moderator variables.*

4. Suppose you want to combine a set of studies, but studies stem from three different countries. Are you going to account for the country, why (not), and if so, how?

A possible answer:

- *it may be a good idea to take the country into account, since the country may be a moderator variable*
- *a possibility is to perform a weighted ANOVA, or to perform a ANOVA if a homogeneity test reveals that study population effects are indeed probably varying*
- *note that if the number of countries would have been larger, a three level meta-analysis was an interesting alternative*